

UNIX Programming & Reference Guide

Imp command for Unix

Command	Strict UNIX?	Working / Meaning	Example
abs	✗ No	Absolute value function	abs (-10)
log	✗ No	Natural logarithm function	log(10)
sin	✗ No	Sine function	sin(30)
cos	✗ No	Cosine function	cos(60)
sqrt	✗ No	Square root function	sqrt(25)
tan	✗ No	Tangent function	tan(45)
reset	⚠ Depends	Reset terminal settings	reset
top	✗ No	Display running processes dynamically	top
tty	✓ Yes	Display current terminal name	tty
umask	✓ Yes	Set/display default permissions	umask 022
uname	✓ Yes	Display system information	uname
who	✓ Yes	Display logged-in users	who
chgrp	✓ Yes	Change file group ownership	chgrp staff file.txt
chown	✓ Yes	Change file ownership	chown indra file.txt
date	✓ Yes	Display date and time	date
echo	✓ Yes	Display text	echo "Hello UNIX"
exit	✓ Yes	Exit current shell	exit
groupadd	✗ No	Create a group	groupadd students
groups	⚠ Depends	Display group membership	groups
kill	✓ Yes	Send signal to a process	kill 1234
logout	✓ Yes	Log out from session	logout
passwd	✓ Yes	Change password	passwd
ps	✓ Yes	Display process status	ps

Command	Strict UNIX?	Working / Meaning	Example
nm	✔ Yes	Display symbol table	nm prog.o
size	✔ Yes	Display executable size	size prog.o
bc	⚠ Depends	Calculator utility	echo "5+3" bc
mysql	✗ No	MySQL database client	mysql -u root -p
write	✔ Yes	Send message to another user	write user1
cat	✔ Yes	Display file contents	cat file.txt
cd	✔ Yes	Change directory	cd /home/indra
chmod	✔ Yes	Change permissions	chmod 755 file.sh
cp	✔ Yes	Copy files	cp file1 file2
file	⚠ Depends	Determine file type	file test.txt
find	✔ Yes	Search files	find . -name "*.txt"
grep	✔ Yes	Search text patterns	grep "UNIX" notes.txt
head	⚠ Depends	Display first lines of file	head file.txt
ln	✔ Yes	Create links	ln file1 file2
ls	✔ Yes	List files/directories	ls -l
mkdir	✔ Yes	Create directory	mkdir project
more	✔ Yes	Display file page by page	more file.txt
mv	✔ Yes	Move/Rename files	mv old.txt new.txt
rename	✗ No	Rename files	rename old.txt new.txt
nice	✔ Yes	Run process with modified priority	nice sort data.txt
rm	✔ Yes	Remove files	rm file.txt

Imp command for Unix (Pattern Matching)

1. Check if File has .sh Extension

```
echo "Enter File Name:"
read file

case "$file" in
    *.sh) echo "Shell Script File" ;;
    *) echo "Not a Shell Script File" ;;
```

```
esac
```

```
# Example Input: test.sh -> Output: Shell Script File
```

2. Show Files with 3 Characters Names

```
ls ???
```

```
# Example:
```

```
# Files: abc, xyz, test, hello
```

```
# Output: abc, xyz
```

3. Show Files Starting with a, b, or c

```
ls [abc]*
```

```
# Example:
```

```
# Files: apple, ball, cat, dog, book
```

```
# Output: apple, ball, book, cook
```

Imp programmes for Unix

1. Check Even or Odd Number

```
echo "Enter a number:"
```

```
read n
```

```
r=`expr $n % 2`
```

```
if [ $r -eq 0 ]
```

```
then
```

```
    echo "Even Number"
```

```
else
```

```
    echo "Odd Number"
```

```
fi
```

2. Check Whether a Number is Positive or Not

```
echo "Enter a number:"
```

```
read n
```

```
if [ $n -gt 0 ]
```

```
then
```

```
    echo "Positive Number"
```

```
else
```

```
    echo "Not Positive"
```

```
fi
```

3. Check Whether a Year is Leap Year or Not

```
echo "Enter Year:"
```

```
read y
```

```
r=`expr $y % 4`

if [ $r -eq 0 ]
then
    echo "Leap Year"
else
    echo "Not Leap Year"
fi
```

4. Check Whether a Person is Senior Citizen or Not

```
echo "Enter Age:"
read age

if [ $age -ge 60 ]
then
    echo "Senior Citizen"
else
    echo "Not Senior Citizen"
fi
```

5. Calculate Road Tax

```
echo "Enter Bike Price:"
read p

if [ $p -gt 100000 ]
then
    tax=`expr $p \* 15 / 100`
elif [ $p -gt 50000 ]
then
    tax=`expr $p \* 10 / 100`
else
    tax=`expr $p \* 5 / 100`
fi

echo "Road Tax = Rs. $tax"
```

6. Calculate Electric Bill

```
echo "Enter Units:"
read u

if [ $u -le 100 ]
then
    bill=0
elif [ $u -le 200 ]
then
    x=`expr $u - 100`
    bill=`expr $x \* 5`
else
    x=`expr $u - 200`
    bill=`expr 100 \* 5 + $x \* 10`
fi
```

```
echo "Bill Amount = Rs. $bill"
```

7. Check Whether a Number is Three Digit or Not

```
echo "Enter Number:"
read n

if [ $n -ge 100 -a $n -le 999 ]
then
    echo "Three Digit Number"
else
    echo "Not Three Digit Number"
fi
```

8. Menu Based Calculator

```
echo "Enter Two Numbers:"
read a b

echo "1.Add"
echo "2.Subtract"
echo "3.Multiply"
echo "4.Divide"

read ch

case $ch in
1) echo `expr $a + $b` ;;
2) echo `expr $a - $b` ;;
3) echo `expr $a \* $b` ;;
4) echo `expr $a / $b` ;;
*) echo "Invalid Choice" ;;
esac
```

9. Sum of First N Natural Numbers

```
echo "Enter N:"
read n

i=1
sum=0

while [ $i -le $n ]
do
    sum=`expr $sum + $i`
    i=`expr $i + 1`
done

echo "Sum = $sum"
```

10. Check Palindrome Number

```
echo "Enter Number:"
read n

temp=$n
rev=0

while [ $n -gt 0 ]
do
    d=`expr $n % 10`
    rev=`expr $rev \* 10 + $d`
    n=`expr $n / 10`
done

if [ $temp -eq $rev ]
then
    echo "Palindrome"
else
    echo "Not Palindrome"
fi
```

11. Check Armstrong Number (3 Digit)

```
echo "Enter Number:"
read n

temp=$n
sum=0

while [ $n -gt 0 ]
do
    d=`expr $n % 10`
    sum=`expr $sum + $d \* $d \* $d`
    n=`expr $n / 10`
done

if [ $temp -eq $sum ]
then
    echo "Armstrong Number"
else
    echo "Not Armstrong Number"
fi
```

12. Count Total Even and Odd Digits

```
echo "Enter Number:"
read n

even=0
odd=0

while [ $n -gt 0 ]
do
    d=`expr $n % 10`
    r=`expr $d % 2`
```

```
if [ $r -eq 0 ]
then
    even=`expr $even + 1`
else
    odd=`expr $odd + 1`
fi
n=`expr $n / 10`
done

echo "Even Digits = $even"
echo "Odd Digits = $odd"
```

UNIX File System Architecture (Theory)

Memory Trick:

Directory = Name mapping | Inode = File Metadata | Disk Block = Actual Data

Directory	Stores Filename and Inode Number link.
Inode	Stores File Type, Owner, Permissions, Size, Timestamps, Data Block locations. (No filename).
Disk Blocks	Stores the actual content bits of the file.

Complete Flow:

Directory → Filename → Inode → Data Blocks → Actual File Data